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Unraveling support effects in Ru-based catalysts for efficient conversion of marine biomass to sugar alcohols

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ABSTRACT

The valorization of marine biomass into high-value chemicals presents both ecological and economic opportunities. This work systematically investigated support-dependent catalytic behavior in ruthenium-based systems for converting *Enteromorpha prolifera* into sugar alcohols (rhamnitol, sorbitol and xylitol). Through comparative analysis of self-synthesized Ru/TS-1 with Ru/SiO_2, Ru/TiO_2 and commercial Ru catalysts (Ru/C and Ru/Al_2O_3), we demonstrated that TS-1 supported catalyst exhibit superior performance through three synergistic mechanisms: (1) enhanced metal-support interactions (MSI) yielding optimal Ru^{3+}/Ru^0 balance (XPS quantification), (2) maximized metallic surface area (78.3 m^2/g) with minimum Ru particle size, and (3) tailored acid-base bifunctionality (NH_3/CO_2-TPD characterization). The optimized Ru/TS-1 achieved exceptional sugar alcohol yields (90–95%, productivity=19.1 $mol_{rhamnitol}$ mol_{Ru}^{-1} h^{-1}) with favorable kinetics (TOF=425.8 h^{-1}, E_a=21.8 kJ·mol^{-1}). Combined experimental and density functional theory (DFT) analyses revealed that moderate acidity/basicity facilitated ring-opening of cyclic rhamnose while preventing excessive cleavage of C3-C4 bond (responsible for acetol formation). Mechanistic studies identified Ru^{3+} sites as critical for substrate adsorption and Ru^0 domains for H_2 activation. This fundamental understanding of support effects provides a rational design strategy for developing advanced catalysts in marine biomass upgrading.

1. Introduction

The sustainable valorization of marine biomass into value-added chemicals has emerged as a dual-resolution strategy addressing both environmental remediation and renewable energy demands (Li et al., 2024; Zhang et al., 2023; Guo et al., 2024; Kang et al., 2024). Among various biomass-based products, sugar alcohols (e.g., sorbitol, xylitol, rhamnitol) have garnered significant attention as versatile platform chemicals. These polyols serve as multifunctional intermediates in food additives, pharmaceutical excipients, and oral care products due to their non-cariogenic properties and biocompatibility (Zada et al., 2017; Zhang et al., 2020; Ullah et al., 2022). Current industrial production predominantly relies on catalytic hydrogenation of monosaccharides using Raney Ni catalysts, yet suffers from severe metal leaching and harsh operating conditions (elevated temperatures >120 °C, H_2 pressures >4 MPa) (Lee et al., 2005). This technological limitation has driven extensive exploration of noble metal alternatives, particularly ruthenium-based systems, owing to their superior hydrogenation activity and structural stability (Guo et al., 2014; Ban et al., 2017).

Recent advances elucidated the critical role of catalyst design in governing reaction pathways. Guo et al. demonstrated 98.8 % sorbitol yield from glucose hydrogenation over Ru/ZSM-5 under moderate conditions (120 °C, 4 MPa H_2), while Ban et al. revealed that Ru-Cu bimetallic interactions could enhance hydrogen spillover effects in alginate conversion (Guo et al., 2014; Ban et al., 2017). Furthermore, Zhang's work on Ru/MWCNTs established a structure-activity relationship between Ru valence states (Ru^0 for H_2 activation vs. Ru^{3+} for substrate adsorption) and catalytic performance (Zhang et al., 2020). Despite these advances, two fundamental challenges persist: (i) uncontrolled C-C cleavage under conventional reaction conditions generates undesired byproducts (e.g., acetol), and (ii) most studies focus on terrestrial biomass-derived hexoses/pentoses, neglecting marine

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polysaccharide resources (Beine et al., 2022; Xiao et al., 2023).

It is generally accepted that the transformation of sugars into sugar alcohols involves the ring-opening (cyclic sugar to chain sugar) and hydrogenation of the aldehyde group to a hydroxyl group. However, the cleavage of C-C bonds in sugars produces side products in the liquid phase under high temperatures and H_2 pressure (Beine et al., 2022; Xiao et al., 2023). Therefore, designing nano catalysts with multifunctional properties to cooperatively catalyze the cascade reaction in biomass transformation towards sugar alcohols production while inhibiting the considerable side-reactions is of great significance and highly desirable.

Enteromorpha prolifera (*E. prolifera*), a proliferative green tide algae in China's Yellow Sea, represents an underutilized marine biomass containing ~45 wt% polysaccharides (rhamnose/glucose/xylose) (Ray, 2006; Zhang et al., 2019). Current utilization strategies predominantly target low-value applications, with limited reports on its conversion to specialty chemicals (Xiao et al., 2023; Xiao et al., 2025). This knowledge gap, coupled with urgent needs for marine ecological remediation, motivates our investigation into catalytic upgrading of *E. prolifera*-derived sugars.

Herein, we presented a systematic exploration of support effects in Ru-based nano catalysts for selective sugar alcohols production from marine biomass. Through comparative evaluation of TS-1 against conventional supports (SiO_2, TiO_2, Al_2O_3, C), we identify three critical design principles: (1) Electronic modulation: TS-1 induced optimal Ru^{3+}/Ru^0 balance via Metal-Support Interactions (MSI), as evidenced by XPS and H_2-TPR. (2) Structural advantages: Ru/TS-1 achieved maximum metallic dispersion and accessible metal surface area (78.3 m^2/g) through confined growth in zeolitic channels. (3) Acid-base synergy: Moderate bifunctionality (NH_3/CO_2-TPD) enabled sequential selective hydrogenation while suppressing C3-C4 cleavage.

The optimized catalyst achieved sugar alcohol yields (90–95 %) from both model compounds and raw *E. prolifera*, with exceptional productivity (19.1 $mol_{rhamnitol}$ mol_{Ru}^{-1} h^{-1}) and energy efficiency (E_a=21.8 $kJ{\cdot}mol^{-1}$). Mechanistic studies combining DFT calculations elucidated how TS-1's framework topology directed reaction pathways: (i) Moderate acid/base sites facilitated cyclic-rhamnose ring-opening, while against the cleavage of C3-C4 bonds. (ii) MSI induced optimal Ru^{3+}/Ru^0, and Ru^{3+} species was contributed to adsorb the substrate, while Ru^0 species was responsible for the activation of H_2. This work established a paradigm for marine biomass valorization through rational catalyst design, bridging critical gaps between heterogeneous catalysis and blue biotechnology.

2. Experimental sections

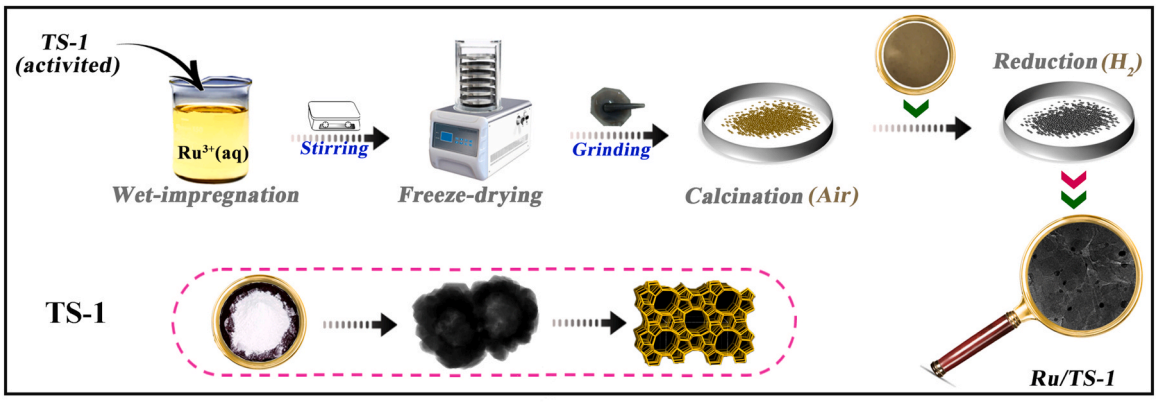
2.1. Catalyst preparation

The Ru-based catalysts were synthesized through a freeze-drying-assisted wet impregnation approach (Scheme 1). In a representative procedure for 3-Ru/TS-1 preparation, pre-calcined TS-1 support (1.0 g, activated at 350 °C for 3 h) was uniformly dispersed in an aqueous $RuCl_3{\cdot}xH_2O$ solution (80.0 mg precursor) under ultrasonic treatment (40 kHz, 30 min), followed by continuous magnetic stirring for 4 h at ambient temperature. The resulting suspension underwent 24 h aging to ensure complete precursor-support interaction prior to solvent removal through lyophilization (-50 °C, 0.1 mbar, 48 h). The freeze-dried intermediate was subsequently pulverized and thermally treated in static air (350 °C for 3 h, 2 °C/min ramp) to decompose chloride species, followed by in situ reduction under H_2 flow (50 mL/min, 350 °C for 2 h, 2 °C/min heating) to generate metallic Ru nanoparticles. The final catalyst was stored under vacuum (0.01 Pa) to prevent surface oxidation. Catalyst nomenclature follows x-Ru/support convention, where x represents the nominal Ru loading (wt%) calculated by the weight of Ru to the weight of support. For comparative studies, control catalysts including Ru/TiO_2 and Ru/SiO_2 were prepared using analogous procedures, while commercial Ru/C and Ru/Al_2O_3 (3 wt% Ru, Sigma-Aldrich and Alfa) served as reference materials. Metal loadings were verified through inductively coupled plasma atomic emission spectrometry (ICP-AES), showing < 5 % deviation from theoretical values.

2.2. Catalytic reaction

Hydrogenation experiments were conducted in a 100 mL stainless-steel batch reactor (Parr Instruments) equipped with a mechanical stirrer and temperature-pressure monitoring systems. In a standard protocol, an aqueous suspension containing rhamnose (200 mg, Admas-Beta, ≥98.5 %), catalyst (100 mg), and deionized water (50 mL) was charged into the reactor. The system underwent three cycles of H_2 purging (99.999 %) to evacuate residual air, followed by pressurization to 3.0 MPa H_2 at room temperature. The reaction mixture was heated to 140 °C (2.5 °C/min ramp) under vigorous stirring (400 rpm) to ensure efficient gas-liquid-solid contact and maintained isothermally for 2 h. After reaction, the reactor was rapidly cooled in an ice bath to quench the reaction. The liquid phase was isolated via vacuum filtration through a 0.22 μm PTFE membrane to remove catalyst particles.

Hydrolysis of *E. prolifera* to carbohydrates was conducted in a 600 mL stainless-steel parr. In a typical run (Scheme S1), 5 g *E. prolifera*, 10 mmol common and cheap of organic acid (e.g. oxalic acid) and 200 mL water were loaded in a 600 mL stainless-steel parr with a mechanical stirrer and temperature-pressure monitoring systems. The reaction mixture was heated to 160 °C under vigorous stirring (600 rpm) for 60 min. After reaction, the solution was purified by activated carbon, dichloromethane (DCM), $Ca(OH)_2$ and ion-exchange resins (Amberlite IRA-67). The concentrated sugars solution was further collected and used for hydrogenation over Ru/TS-1 catalyst.



Scheme 1. Schematics of the preparation of Ru/TS-1 catalyst.

2.3. Product analysis

Quantitative analysis of reaction products (acetol, rhamnitol, sorbitol and xylitol, etc.) were performed by high-performance liquid chromatography (HPLC, Agilent 1260 Infinity II) equipped with a refractive index detector, using a Bio-Rad Aminex HPX-87H column maintained at 50 °C (5 mM H_2SO_4 mobile phase, 0.6 mL/min flow rate).

The conversion (Con.) of starting material, product yield (Yield), productivity (Prod.), turnover frequency (TOF) and dispersion of Ru (D_{Ru}) were calculated by the Eqs. (1), (2), (3), (4) and (5), respectively:

$$\text{Con.}(\%) = \frac{\text{moles of initial substrate} - \text{moles of remained substrate}}{\text{moles of initial substrate}} \times 100\% \quad (1)$$

$$\text{Yield}(\text{C} - \text{mol}\%) = \frac{\text{moles of carbon in product}}{\text{moles of carbon in feedstock}} \times 100\% \quad (2)$$

$$\text{Prod.}\ (\text{mol}_{\text{rhamnitol}}\ \text{mol}_{\text{Ru}}^{-1}\ \text{h}^{-1}) = \frac{\text{moles of rhamnitol}}{\text{moles of Ru} * t} \quad (3)$$

$$\text{TOF}(\text{h}^{-1}) = \frac{\text{moles of initial substrae} - \text{moles of remained subsrate}}{\text{moles of Ru} * D_{Ru} * t} \quad (4)$$

$$\text{D}_{\text{Ru}} = (2 \times \text{H}_2\text{uptake} \times \text{M}_{\text{Ru}})/[\text{Ru}]_{\text{loading}} \quad (5)$$

The productivity was calculated based on the references (Han et al., 2016; Zhang et al., 2023). The *t* in Eq. (3) meant the reaction time (2 h), and moles of Ru in (3) and (4) represented the actual moles of Ru loaded on the catalysts determined by ICP-AES. The *t* in Eq. (4) meant the reaction time (10 min), and D_{Ru} in Eq. (5) represented the Ru dispersion of different Ru-based catalysts estimated by H_2 chemisorption results, where H_2 uptake was the moles of H_2 adsorbed on per gram catalyst, M_{Ru} was the molecular weight of Ru (101.1 g/mol), [Ru] loading was the actual loading of Ru on catalyst determined by ICP-AES.

All the details of materials, catalyst characterizations and DFT calculations were summarized in the Supporting Information.

3. Results and discussion

3.1. Catalytic hydrogenation of rhamnose

Initial control experiments under H_2 atmosphere without catalysts yielded minimal rhamnose conversion (3.1 %, Fig. 1a), producing trace of rhamnitol and acetol-a result consistent with prior work (Xiao et al., (2023)). Catalytic testing revealed significant enhancement using commercial systems: Ru/C achieved 27.6 % rhamnitol yield at 46.1 % substrate conversion, while Ru/Al_2O_3 demonstrated higher activity (91.5 % conversion) but lower selectivity, generating substantial acetol (25.2 %, Fig. 1a) alongside 41.2 % rhamnitol.

Systematic evaluation of synthesized Ru catalysts supported on metal oxides/zeolites uncovered pronounced support effects. Ru/SiO_2 and Ru/TiO_2 exhibited moderate performance, delivering rhamnitol yields of 48.1 % and 60.9 %, respectively. Remarkably, the Ru/TS-1 system displayed exceptional catalytic efficiency, achieving 87.7 % rhamnitol yield at near-complete substrate conversion (98.2 %, Fig. 1a). This hierarchical activity trend (Ru/TS-1 > Ru/TiO_2 > Ru/SiO_2) underscored the critical role of support properties in modulating catalyst-substrate interactions and intermediate stabilization.

Given the best catalytic activity of Ru/TS-1 catalyst, we then optimized the reaction conditions for rhamnose transformation to rhamnitol over Ru/TS-1 catalyst, including Ru loading, reaction temperature, reaction time, catalyst/substrate ratio and H_2 pressure (Fig. 1b and Fig. 2). It was indicated that rhamnitol yield was only 3.8 % over TS-1 catalyst (Table S2, entry 1), and which was gradually increased with the increment of Ru loading, and got the maximum at 3 wt%. Continuously increasing Ru loading failed to further enhance rhamnitol yield (Fig. 1b), but slightly raised the yield of acetol, possibly due to the fact that high Ru loading facilitated the cleavage of C-C bond of rhamnose in H_2 atmosphere (Buta et al., 2025; Lazaridis et al., 2017).

Rising reaction temperature obviously enhanced acetol yield, and the maximum rhamnitol yield was obtained at 150 ℃ (Fig. 2a). At this temperature, rhamnitol yield gradually increased with prolonging reaction time, and the highest value was obtained at 2 h (Fig. 2b). However, higher temperature (160 ℃) and longer reaction time (2.5 h) led to slight decrement in rhamnitol yield. The rhamnitol yield was increased with the increment of catalyst/substrate ratio, and got the maximum at the ratio of 1:2 (Fig. 2c). H_2 pressure showed a remarkable influence on

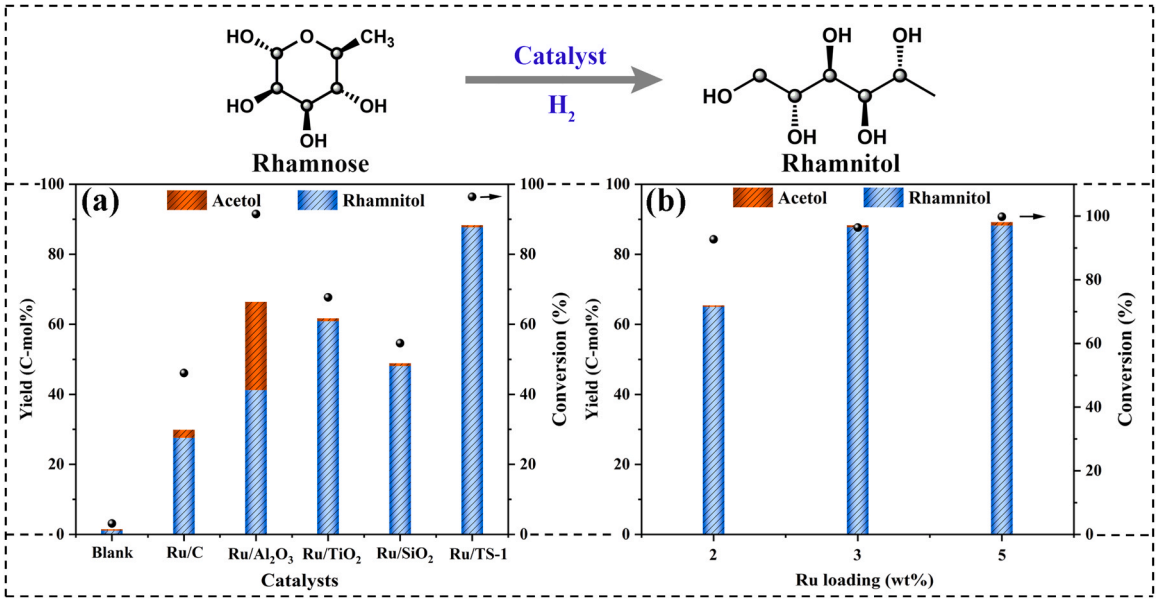


Fig. 1. The catalytic activity of various Ru-based catalysts (a) and the effect of Ru loading (b) of Ru/TS-1 catalyst on the conversion of rhamnose to rhamnitol. Reaction conditions: 0.2 g rhamnose, 0.1 g catalyst (Ru loading was 3 wt% if metal catalyst was used), 50 mL water, 140 ℃, 3 MPa H_2, 2 h.

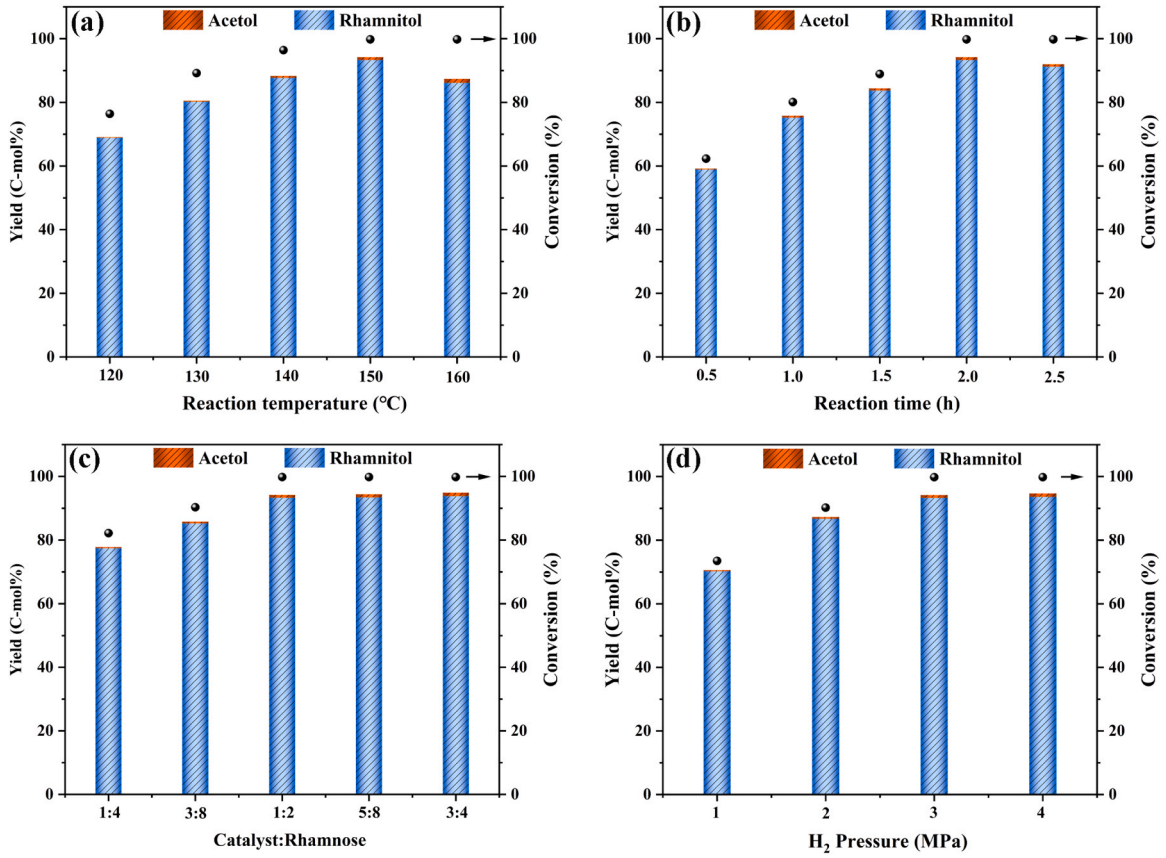
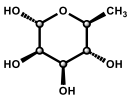
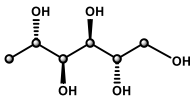
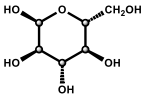
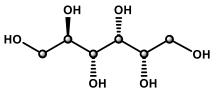
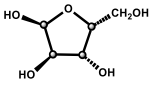
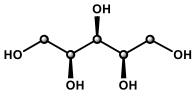


Fig. 2. The effect of reaction temperature (a), reaction time (b), the ratio of catalyst to substrate (c) and H_2 pressure (d) on the conversion of rhamnose to rhamnitol. Reaction conditions: 0.2 g rhamnose, 0.1 g catalyst ((a, b, d) 3-Ru/TS-1), 50 mL water, 150 ℃ (b, c, d), 2 h (a, b, d), 3 MPa H_2 (a, b, c).

the conversion of rhamnose. There was no rhamnitol formation in N_2 atmosphere (Table S2, entry 2). A minor increase of H_2 pressure caused a sharp increment in rhamnitol yield, and the maximum of rhamnitol yield was obtained at 3 MPa H_2 pressure. Further increasing H_2 pressure led to no obvious increment rhamnitol yield (Fig. 2d). On the basis of these results, it was speculated that the formation of acetol and rhamnitol were two competing reactions, wherein high Ru loading, reaction temperature and reaction time could promote the cleavage of C3-C4 bond in rhamnose to yield acetol. Under the optimal reaction conditions (150 ℃, 2 h, 3 MPa H_2, 3-Ru/TS-1:Rhamnose =1:2), the maximum yield of rhamnitol was up to 93.3 % (Productivity=19.1 $mol_{rhamnitol}$ mol_{Ru}^{-1} h^{-1}, Table S3). Furthermore, ESI-MS patterns of reaction solution proved the existence of rhamnitol (Fig. S1), and NMR patterns disclosed the high rhamnitol purity in the reaction solution (Fig. S2).

To evaluate practical applicability, *Enteromorpha prolifera* biomass was processed through an integrated biorefinery approach. Primary hydrolysis screening identified oxalic acid (0.5 M, 160 °C, 1 h) as the optimal catalyst among readily available, cost-effective organic acids (Fig. S3), achieving maximal saccharification efficiency with predominant release of rhamnose (94.0 mg/g biomass), glucose (32.9 mg/g), and xylose (39.2 mg/g). The acid hydrolysate underwent sequential purification via ion-exchange resins (Amberlite IRA-67) and activated

Table 1
The hydrogenation of various *Enteromorpha prolifera* derived sugars to sugar alcohol over 3-Ru/TS-1 catalyst.

Entry	Substrate		Product		Yield (C-mol%)
1	Rhamnose[a]		rhamnitol		93.3
2	Rhamnose[b]		rhamnitol		90.9
3	Glucose[a]		Sorbitol		94.7
4	Glucose[b]		Sorbitol		92.0
5	Xylose[a]		Xylitol		96.6
6	Xylose[b]		Xylitol		93.8

Reaction conditions: 0.2 g substrate, 0.1 g 3-Ru/TS-1, 150 ℃, 50 mL water, 3 MPa H_2, 2 h.
[a] Model compounds
[b] *Enteromorpha prolifera* derived (oxalic acid)

carbon adsorption (Zhao et al., 2019; Zhou et al., 2020), yielding a concentrated sugar stream (≈120 g/L total sugars) suitable for catalytic upgrading.

Notably, the 3-Ru/TS-1 system maintained exceptional performance with real biomass-derived substrates, achieving ≥ 90 % combined sugar alcohol yield (Table 1) under aqueous-phase hydrogenation conditions (150 °C, 3 MPa H_2, 2 h). Comparative analysis revealed < 5 % yield variance between model sugars and biomass-derived counterparts, demonstrating the catalyst's robustness against complex feedstock matrices.

3.2. Kinetic study

Temperature-dependent kinetic analysis (Fig. 3, Table 2) revealed a pronounced enhancement in rhamnose hydrogenation efficiency, with rhamnitol yield increasing significantly as the reaction temperature rose from 403 K to 423 K. Assuming pseudo-first-order kinetics, as established in prior studies (Gao et al., 2022), apparent activation energies (E_a) for the Ru catalysts supported on TS-1, TiO_2, and SiO_2 were determined through linear regression of Arrhenius plots. The Ru/TS-1 system exhibited the lowest E_a (21.8 kJ·mol^{-1}), underscoring its superior catalytic activity, while Ru/SiO_2 displayed the highest energy barrier (E_a=36.0 kJ·mol^{-1}). Furthermore, Ru/TS-1 achieved a remarkable turnover frequency (TOF) of 425.8 h^{-1} at 423 K (Table S4).

3.3. Catalyst characterization

The exceptional catalytic performance of Ru/TS-1 in rhamnose-to-rhamnitol conversion was investigated through systematic characterization of synthesized Ru-based catalysts (3 wt% loading) supported on different supports. N_2 adsorption-desorption analysis (Fig. S4) revealed that all support materials and corresponding Ru-loaded catalysts exhibited type IV adsorption isotherms with H1-type hysteresis loops, characteristic of mesoporous structures. Pore size distribution profiles (Fig. S5) demonstrated predominantly mesoscale dimensions ranging from 2 to 30 nm. As detailed in Table S5, TS-1 support displayed superior textural properties with the highest specific surface area (258 m^2/g) and largest pore volume (0.22 cm^3/g), followed by TiO_2 and SiO_2. Catalyst preparation through Ru impregnation caused only marginal

Table 2
The hydrogenation of rhamnose to rhamnitol over different catalysts at different temperatures.

Catalyst	Temp. (K)	k (min^{-1})	E_a (kJ·mol^{-1})
Ru/TS−1	403	0.026	21.8
	413	0.031	
	423	0.035	
Ru/TiO_2	403	0.020	28.7
	413	0.025	
	423	0.030	
Ru/SiO_2	403	0.014	36.0
	413	0.020	
	423	0.026	

Temp. means reaction temperature. k is the reaction rate (kinetic) constant for 100 mg catalyst. Reaction conditions: 0.2 g rhamnose, 0.1 g catalysts, 50 mL H_2O, 3 MPa H_2.

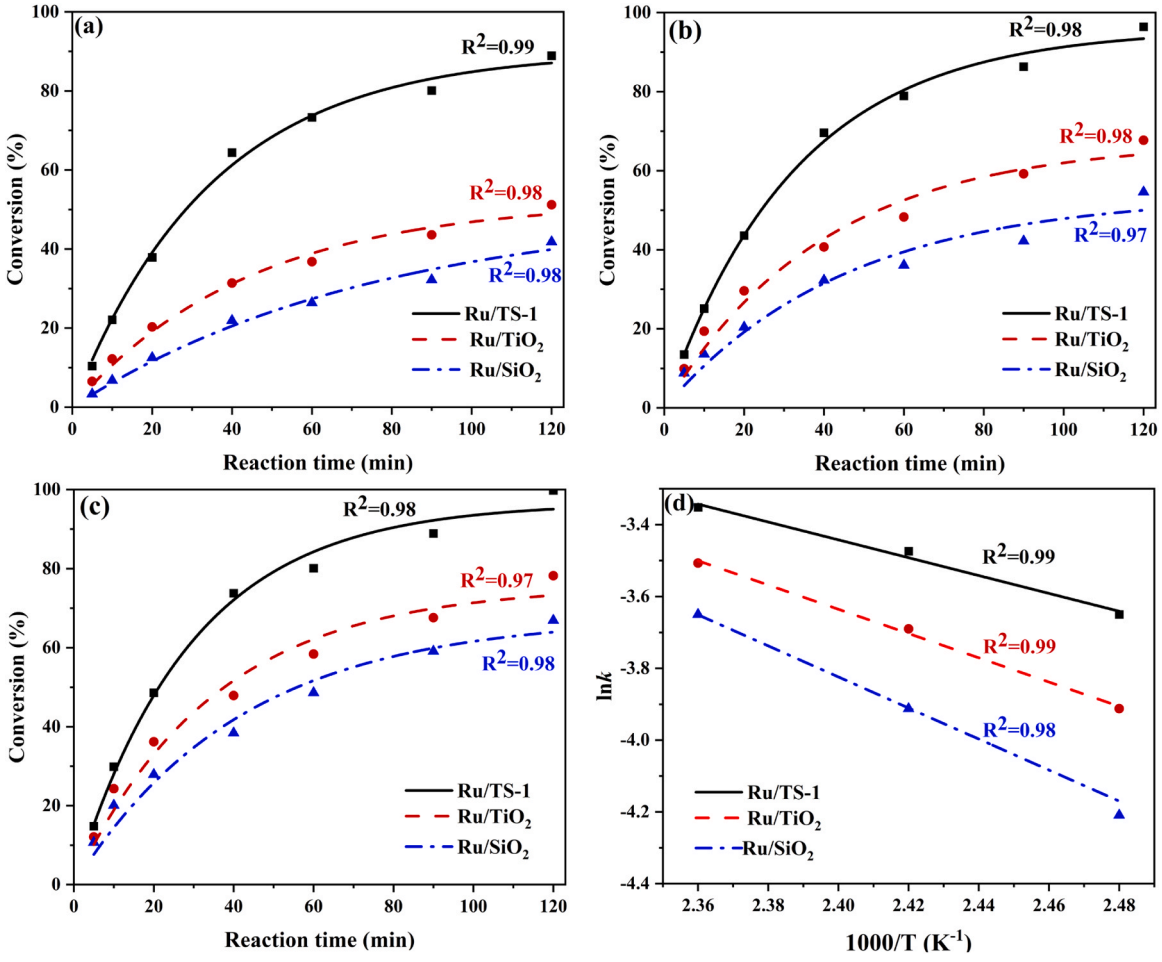


Fig. 3. The catalytic conversion of rhamnose to rhamnitol over different catalysts at 403 K (a), 413 K (b), 423 K (c) and the linear relation between ln$k$ versus 1000/T (d). Reaction conditions: 0.2 g rhamnose, 0.1 g catalyst, 50 mL water, 3 MPa H_2.

reductions in both surface area and pore volume across all supports (Table 3), indicating effective preservation of the original mesoporous architecture during metal loading. XRD patterns (Fig. S6) revealed distinct diffraction peaks corresponding to the crystalline frameworks of the parent materials, confirming that Ru incorporation did not alter the structural integrity of the supports. Notably absent were characteristic peaks for metallic Ru species (2θ≈38° and 44°), suggesting high dispersion of Ru nanoparticles on the supports (Tamura et al., 2022; Machado et al., 2022).

ICP-AES analysis verified high metal loading efficiencies (>90 %) across all catalysts (Table S6), in well agreement with the result of SEM-EDS (Fig. 4a). SEM-Mapping of 3-Ru/TS-1 catalyst (Fig. 4b) showed that Ru species were uniformly dispersed on the surface of TS-1 support. TEM images demonstrated that the average particle size of Ru species on 3-Ru/TS-1, 3-Ru/TiO_2 and 3-Ru/SiO_2 were 6.2 ± 1.5 nm, 8.5 ± 1.8 nm, and 8.8 ± 2.0 nm, respectively. This implied that TS-1 enabled the best dispersion of Ru species, due to the highest surface area and largest pore volume. Moreover, HRTEM images (Figs. 4c-2) indicated that all as-prepared Ru-based catalysts showed a crystal spacing of 0.20 nm, which was assigned to metallic Ru (101) planes.

The H_2-TPR experiment revealed distinct reduction characteristics of supported Ru precursors. All catalysts exhibited primary reduction events between 125 and 150°C, corresponding to the reduction of Ru oxide species. Notably, these temperatures significantly exceeded the reduction threshold of unsupported Ru oxides (100°C) (Gopakumar et al., 2023; Yan et al., 2019), demonstrating support-mediated the interaction between Ru species and supports. Particularly, the TS-1-supported catalyst displayed an exceptional high-temperature reduction peak at 227 °C, indicating obvious electron transfer between Ru species and the TS-1 framework. This strong metal-support interaction (SMSI) correlates with TS-1's superior textural properties (surface area: 258 m^2/g; pore volume: 0.22 cm^3/g) observed in physisorption analysis, suggesting structural confinement effects within the mesoporous architecture facilitate enhanced interfacial bonding.

The XPS spectra of various Ru-based catalysts showed obvious Ru 3p peaks (Fig. 5b), which could be deconvoluted to two peaks at ~462.0 eV and ~465.6 eV, assigning to Ru^0 and Ru^{3+}, respectively (Liu et al., 2024; Tabassum et al., 2022). Quantitative analysis demonstrated Ru^0 was the major species in all the cases, and Ru/TS-1 gave the highest Ru^0 content (85.4 %) among the as-prepared Ru-based catalysts, followed by Ru/TiO_2 (76.5 %) and Ru/SiO_2 (70.1 %). Moreover, the O 1 s spectra (Fig. 5c) could be deconvoluted to two peaks, corresponding to two oxygen species: O_I, oxygen species chemisorbed at oxygen vacancies at ~531.1 eV, which could reflect the concentration of surface oxygen vacancies; O_{II}, lattice oxygen at ~529.3 eV (Zhou et al., 2022; Feng et al., 2025; Zhang, et al., 2024). It was found that Ru/TS-1 exhibited the highest concentration of oxygen vacancies (40.7 %), followed by Ru/TiO_2 (26.6 %) and Ru/SiO_2 (11.6 %), which might promote the hydrogenation reaction (Zeng et al., 2023; Araújo1 et al., 2022).

The acidic and basic properties of the as-prepared Ru catalysts were systematically investigated through temperature-programmed desorption (TPD) analyses using NH_3 and CO_2 as probe molecules, respectively. Deconvolution of NH_3-TPD profiles (Fig. 5d) revealed three

distinct acid site populations: weak (<200 °C), medium (200–400 °C), and strong acid sites (>400 °C) (Yan et al., 2021; Xiao et al., 2022).

Table 3
The BET results, metal diameters and metal loading of the prepared catalysts.

Catalysts	S_{BET} (m^2/g)	V_{pore} (cm^3/g)	d_{pore} (nm)	d_{TEM} (nm)	D_{Ru} (%)	Ru loading (wt%)
Ru/TS−1	230	0.20	5.1	6.2	17.1	2.71
Ru/TiO_2	52	0.13	9.3	8.5	13.6	2.69
Ru/SiO_2	31	0.10	4.0	8.8	11.7	2.66

Quantitative analysis demonstrated that Ru/TS-1 possessed the highest concentration of medium-strength acid sites (0.080 mmol/g) and total acid density (0.233 mmol/g), exceeding those of Ru/TiO_2 and Ru/SiO_2 by ~52 % and ~61 %, respectively (Table S7). CO_2-TPD measurements (Fig. S7, Table S8) indicated that Ru/TS-1 exhibited substantially enhanced basicity compared to its counterparts, with Ru/TiO_2 and Ru/SiO_2 showing negligible basic site populations. This distinct acid-base bifunctionality contrasts with commercial references (Figs. S8 and S9, Tables S7 and S8): Ru/C displayed limited surface acidity (0.045 mmol/g) and basicity (0.033 mmol/g), while Ru/Al_2O_3 showed predominant acidity (1.021 mmol/g) and basic characteristics (0.859 mmol/g), aligning with our previous mechanistic studies (Xiao et al., 2023).

3.4. Catalyst performance and reaction mechanism

The hydrogenation activity of rhamnose to rhamnitol exhibited significant dependence on the physicochemical properties of Ru-loaded supports, as evidenced by systematic catalytic evaluations. A synergistic interplay between support and metal characteristics-including surface area, metal-support interaction, metal dispersion, and acid-base functionality governed the Ru loading behavior and subsequent catalytic efficiency. Notably, the TS-1 support outperformed TiO_2 and SiO_2 counterparts, primarily due to its superior surface area (258 m^2/g, Table S5), which promoted Ru nanoparticle dispersion and maximized accessible metal surface area (78.3 m^2/g, Table S9). This structural advantage facilitated optimal reactant-metal interactions, consistent with established principles of supported catalyst design. While a positive correlation between Ru metal surface area and rhamnitol yield was observed for the synthesized catalysts (Table S9), marked deviations emerged when comparing with commercial Ru/C and Ru/Al_2O_3. Notably, Ru/TS-1 achieved a superior rhamnitol yield of 93.3 % at 78.3 $m^2 \cdot g^{-1}$ Ru surface area, whereas commercial Ru/C and Ru/Al_2O_3 exhibited significantly lower activity of 44.5 % and 51.2 % yield despite comparable metal surface area (63.1 and 67.7 $m^2 \cdot g^{-1}$), respectively. This discrepancy intensified in pairwise comparisons: Ru/TiO_2 (65.2 % yield, 62.2 $m^2 \cdot g^{-1}$) outperformed Ru/C (44.5 %, 63.1 $m^2 \cdot g^{-1}$) by 46 % under identical conditions, while Ru/SiO_2 (56.1 % yield, 53.6 $m^2 \cdot g^{-1}$) surpassed Ru/C despite a 15 % lower metal surface area. These findings underscored that catalytic activity for rhamnose-to-rhamnitol conversion was governed not solely by Ru metal surface area but also by synergistic electronic and acid-base interactions. Ru/TS-1, exhibiting optimal density of acid-base bifunctional sites coupled with the highest Ru dispersion, achieved a maximal rhamnitol yield of 93.3 %. In stark contrast, commercial Ru/C, despite comparable Ru surface area (78.3 vs. 63.1 $m^2 \cdot g^{-1}$), yielded only 44.5 % rhamnitol, a deficiency directly correlated with its negligible acid-base functionality (Tables S7-S8). This disparity supports a dual-requirement mechanism: (1) sufficient Ru surface area to activate H_2 and adsorb reactants, and (2) acid-base sites to modulate reaction pathways. As demonstrated in our prior mechanistic studies (Xiao et al., 2023), acid-base pairs in Ru/TS-1 likely facilitated the rate-limiting ring-opening of cyclic rhamnose to its linear form, a prerequisite for selective hydrogenation. However, excessive acid-base strength, as exemplified by Ru/Al_2O_3 (Tables S7-S8), induced undesirable C3-C4 scission of linear intermediates, favoring C_3 byproducts (e.g., acetol) over rhamnitol (Xiao et al., 2023). DFT calculations demonstrated that the adsorption energy of H_2 on Ru^0 (-33.3 kcal·mol^{-1}) was much lower than that on Ru^{3+} (-11.3 kcal·mol^{-1}). In this case, the bond length of H_2 increased from 0.748 to 2.287 Å on Ru^0, suggesting the activation of H_2 to *H by Ru^0 species. Furthermore, the adsorption energy of rhamnose on Ru^{3+} (-20.7 kcal·mol^{-1}) was much lower than that on Ru^0 (-4.8 kcal·mol^{-1}, Figs. 6 and S10), and the C3-C4 bond length of rhamnose exhibited little change (1.523 Å vs 1.526 Å), verifying that Ru^{3+} was more favorable for the adsorption of rhamnose, while have no obvious positive effect on the cleavage of C3-C4 bond in rhamnose (Zhang et al., 2020).

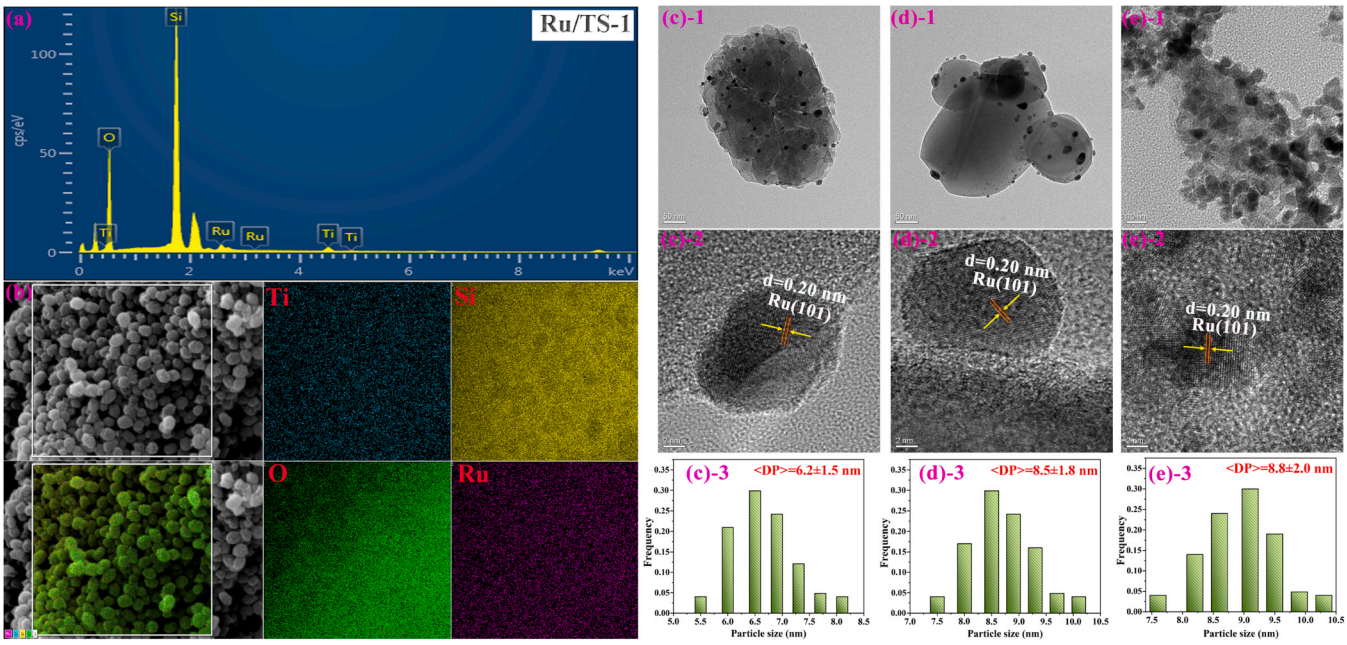


Fig. 4. SEM-EDS patterns (a), SEM-Mapping patterns (b) for 3-Ru/TS-1, TEM images (1), HRTEM images (2) and the distribution of particle size (3) for 3-Ru/TS-1 (c), 3-Ru/TiO_2 (d), 3-Ru/SiO_2 (e) catalysts.

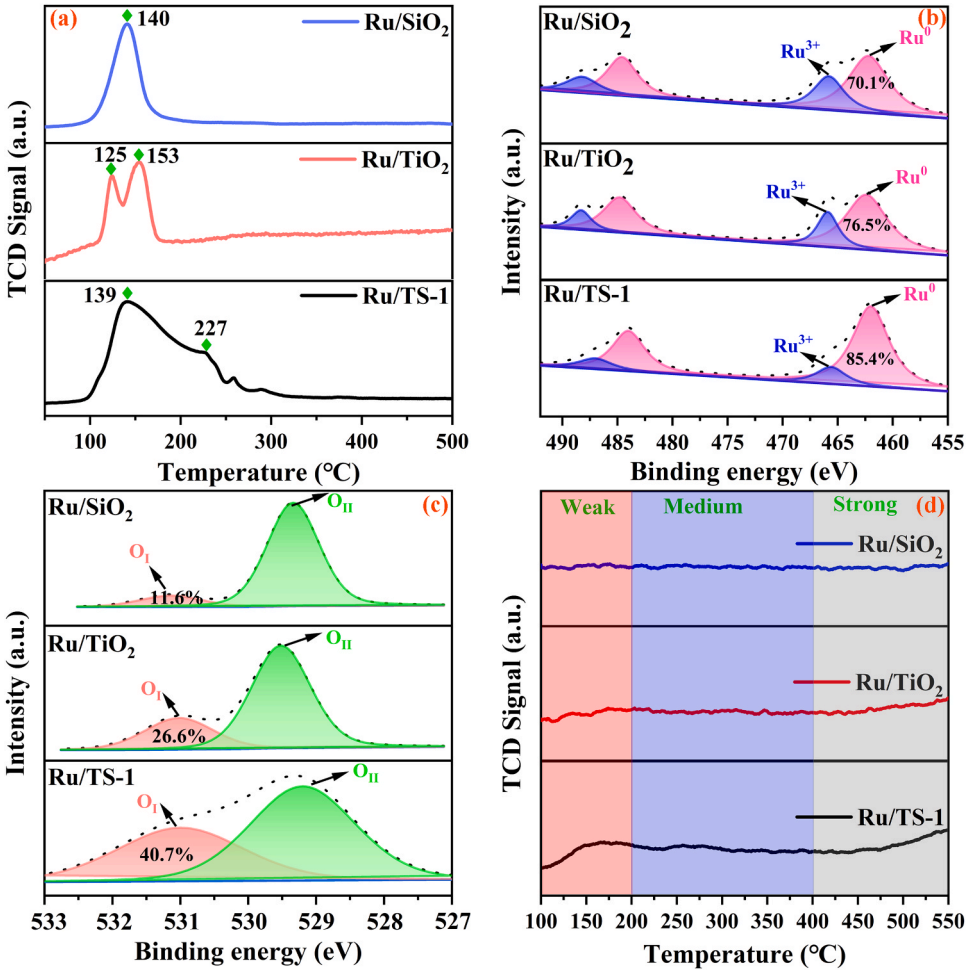


Fig. 5. H_2-TPR patterns (a), XPS spectra of Ru 3p (b), O 1 s (c) and NH_3-TPD patterns (d) for Ru-based catalysts with different supports. Note: The catalysts in H_2-TPR experiment were not reduced by H_2.

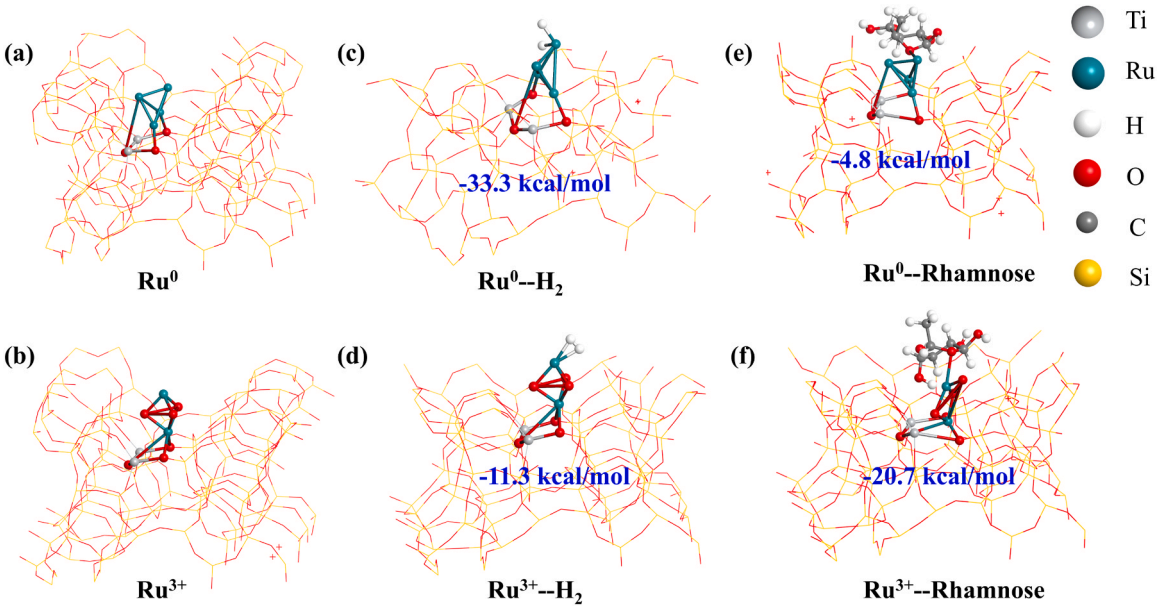


Fig. 6. Ru^0 (a) and Ru^{3+} (b) cluster model in Ru/TS-1 catalyst, and the adsorption energy of H_2 and rhamnose on Ru^0 (c, e) and Ru^{3+} (d, f), respectively.

The superior performance of Ru/TS-1 in rhamnose hydrogenation arises from a synergistic interaction between Ru species and acid-base sites, as delineated in Scheme 2. The reaction proceeds via three concerted steps: (1) Cyclic rhamnose adsorbed preferentially onto oxidized Ru^{3+} sites at the metal-support interface, polarizing the C-O-C bond for subsequent activation. (2) Adjacent acid-base site pairs (identified via NH_3/CO_2-TPD) catalyze the ring-opeing of cyclic rhamnose to its chain form through concerted proton transfer, lowering the energy barrier for ring-opening as demonstrated in our prior DFT studies (Xiao et al., 2023). (3) Metallic Ru^0 nanoparticles dissociatively adsorbed H_2, generating active hydrogen species (*H), and the chain rhamnose intermediate undergoes stepwise *H addition at the carbonyl group, facilitated by electron-deficient Ru sites that suppress undesired C-C cleavage pathways prevalent on strongly acidic/basic supports (e.g., Al_2O_3).

Table 4
The scale-up and stability experiments for the rhamnose hydrogenation over Ru/TS-1 catalyst.[a].

$n_{substrate}$ (mmol)	$m_{catalyst}$ (mg)	Reaction time (h)	Con. (%)	Sel. (%)
1.1	100	2	99.8	93.5
5.5	100	4	40.2	92.8
5.5	200 (1st)[b]	4	98.1	92.7
5.5	200 (2nd)[b]	4	97.4	92.0
5.5	200 (3rd)[b]	4	96.9	92.2
5.5	200 (4th)[b]	4	96.6	92.1

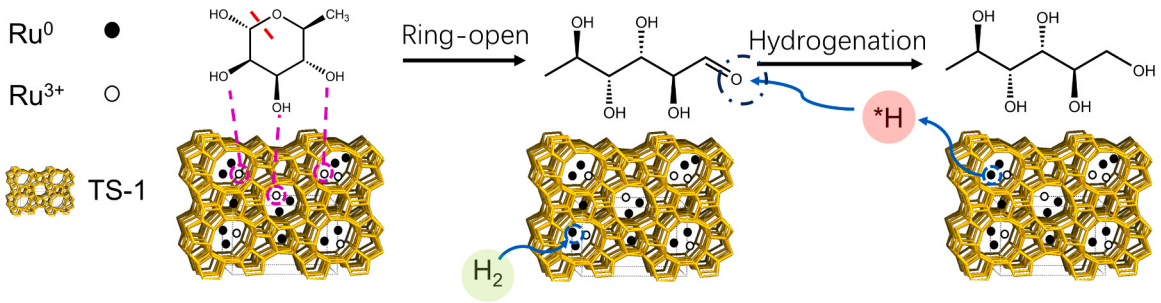
$n_{substrate}$means the moles of the rhamnose, $m_{catalyst}$ means the mass of the 3-Ru/TS-1 catalyst, Con. means the conversion of rhamnose, Sel. means the selectivity of rhamnitol.

[a] Reaction conditions: rhamnose, 50 mL H_2O, 3 MPa H_2, 150 °C.

[b] Catalysts stability experiments.

3.5. Scale-up and stability experiments for the rhamnose hydrogenation over Ru/TS-1 catalyst

To assess the industrial viability of Ru/TS-1 for rhamnose hydrogenation, scalability and catalyst stability were systematically investigated under intensified conditions (Table 4). At elevated substrate concentrations (5.5 mmol, substrate-to-metal ratio = 102.8 $mol_{rhamnose}$ mol_{Ru}^{-1}), the catalyst retained ~93 % selectivity to rhamnitol despite a reduction in conversion (~60 %) relative to dilute systems (1.1 mmol, 41.1 $mol_{rhamnose}$ mol_{Ru}^{-1}). By optimizing catalyst loading (400 mg) and reaction duration (4 h), near-quantitative conversion (98.1 %) with > 92 % selectivity was achieved, demonstrating process scalability without compromising product fidelity. Recycling studies revealed exceptional stability. After four consecutive cycles, the 3-Ru/TS-1 catalyst maintained 94.7 % conversion and 92.2 % selectivity, outperforming conventional sugar hydrogenation catalysts. Structural integrity was confirmed by XRD (Fig. S11), showing no phase degradation or Ru agglomeration post-reaction. ICP-AES analysis further verified negligible Ru leaching (<0.2 wt%), underscoring the robustness of the TS-1 framework in stabilizing active sites. These results position Ru/TS-1 as an industrially relevant catalyst, combining high turnover efficiency, process adaptability, and operational durability-critical



Scheme 2. The proposed pathway of conversion of rhamnose to rhamnitol over Ru/TS-1 catalyst.

attributes for commercial biomass valorization.

4. Conclusions

In this work, we comparatively investigated the catalytic activity of Ru-based catalysts with various supports for the sugar alcohols (e.g. rhamnitol, sorbitol and xylitol) production from biomass waste-*E. profilera*. Ru/TS-1 exhibited the highest catalytic activity, achieving as high as 90–95 % (productivity=19.1 $mol_{rhamnitol}$ mol_{Ru}^{-1} h^{-1}) of sugar alcohol yield, which TOF being 425.8 h^{-1} and E_a as low as 21.8 kJ·mol^{-1}. It was revealed that Ru species and TS-1 support cooperatively contributed to the sugar alcohol formation. Wherein, the most appropriate amount of acid and basic sites benefited for the ring-opening reaction of cyclic-rhamnose to chain-rhamnose, and hindered the further C3-C4 cleavage. Highest metal surface area and Ru^0 species might promote the activation of H_2 and further hydrogenation of sugars to sugar alcohols. Ru/TS-1 also showed great stability (>95 % rhamnose conversion and>92 % rhamnitol yield during third recycling tests) in scale-up experiments. The insights in this work might provide some useful paradigm on the design of catalyst for the conversion of marine biomass into valuable chemicals in a green way.

CRediT authorship contribution statement

Yuan Xiao: Project administration, Funding acquisition, Conceptualization, Investigation, Methodology, Data curation, Validation, Writing – original draft. **Kaizhuo Lv:** Investigation, Methodology, Data curation. **Yuhong Ma:** Investigation, Methodology. **Wanping Zheng:** Methodology, Writing – review & editing. **Zhao Chen:** Investigation, Methodology. **Qiuli Yao:** Supervision, Methodology, Writing – review & editing. **Jinshan Xiong:** Methodology, Funding acquisition, Validation, Formal analysis, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2025.121848.

Data availability

Data will be made available on request.

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